1. Assume we have a dataset $\{(x^{(1)}, y^{(1)}), (x^{(2)}, y^{(2)}), (x^{(3)}, y^{(3)})\} = \{(1,2), (2,4), (3,5)\}$ of points in the 2D plane, and we want to estimate the following linear regression model:

$$\hat{y} = w_1 x + w_0.$$

Compute w_0, w_1 and $\hat{y}^{(1)}$, $\hat{y}^{(2)}$, $\hat{y}^{(3)}$.

$n = 3$ ⟶ avem 3 input-uri (exemple)

$d = 1$ ⟶ features ; dacă $x^{(i)} = (1, 2)$ ⟶ 2 features

$$X = \begin{bmatrix} 1 & 1 \\ 1 & 2 \\ 1 & 3 \end{bmatrix} \in \mathbb{R}^{n \times (d+1)} = \mathbb{R}^{3 \times 2}$$

$$Y = \begin{bmatrix} 2 \\ 4 \\ 5 \end{bmatrix}$$

$$X^T \cdot X = \begin{pmatrix} 1 & 1 & 1 \\ 1 & 2 & 3 \end{pmatrix} \begin{pmatrix} 1 & 1 \\ 1 & 2 \\ 1 & 3 \end{pmatrix} = \begin{pmatrix} 3 & 6 \\ 6 & 14 \end{pmatrix}$$

$\in \mathbb{R}^{2 \times 3}$ $\quad \in \mathbb{R}^{3 \times 2}$ $\quad \longrightarrow \in \mathbb{R}^{2 \times 2}$

$$(X^T X)^{-1} = A^{-1} = \frac{1}{\det A} A^*$$

$$\det A = 3 \cdot 14 - 6 \cdot 6 = 42 - 36 = 6 > 0$$

$$A^T = \begin{pmatrix} 3 & 6 \\ 6 & 14 \end{pmatrix}$$

$$a_{11} = (-1)^{1+1} |14| = 14 \qquad a_{21} = (-1)^{2+1} |6| = -6$$

$$a_{12} = (-1)^{1+2} |6| = -6 \qquad a_{22} = (-1)^{2+2} |3| = 3$$

$$A^* = \begin{pmatrix} 14 & -6 \\ -6 & 3 \end{pmatrix}$$

$$\Rightarrow (X^T X)^{-1} = \frac{1}{6} \begin{pmatrix} 14 & -6 \\ -6 & 3 \end{pmatrix}$$

$$w^* = (X^T \cdot X)^{-1} X^T y =$$

$$= \frac{1}{6} \begin{pmatrix} 14 & -6 \\ -6 & 3 \end{pmatrix} \begin{pmatrix} 1 & 1 & 1 \\ 1 & 2 & 3 \end{pmatrix} \begin{pmatrix} 2 \\ 4 \\ 5 \end{pmatrix} =$$

$$= \frac{1}{6} \begin{pmatrix} 8 & 2 & -4 \\ -3 & 0 & 3 \end{pmatrix} \begin{pmatrix} 2 \\ 4 \\ 5 \end{pmatrix} = \frac{1}{6} \begin{pmatrix} 4 \\ 9 \end{pmatrix} = \begin{pmatrix} \frac{2}{3} \\ \frac{3}{2} \end{pmatrix} \in \mathbb{R}^{2\times 1}$$

w_0 (↗ $\frac{2}{3}$), w_1 (↘ $\frac{3}{2}$)

$$\hat{y} = X w^* = \begin{pmatrix} 1 & 1 \\ 1 & 2 \\ 1 & 3 \end{pmatrix} \begin{pmatrix} \frac{2}{3} \\ \frac{3}{2} \end{pmatrix} = \begin{pmatrix} \frac{13}{6} \\ \frac{22}{6} \\ \frac{31}{6} \end{pmatrix} \in \mathbb{R}^{3\times 1}$$

$\hat{y}^{(1)}$, $\hat{y}^{(2)}$, $\hat{y}^{(3)}$

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Assume we have a dataset $\{(x^{(1)}, y^{(1)}), (x^{(2)}, y^{(2)}), \ldots, (x^{(n)}, y^{(n)})\}$ of points in the 2D plane, and we want to estimate the following linear regression model:

$\hat{y} = w_1 x + w_0$.

Compute w_0, w_1 in terms of $x^{(1)}, y^{(1)}, x^{(2)}, y^{(2)}, \ldots, x^{(n)}, y^{(n)}$. Numerical application: $(1,1), (2,-1), (3,2)$.

$n = 3$

$d = 1$

$$X = \begin{bmatrix} 1 & 1 \\ 1 & 2 \\ 1 & 3 \end{bmatrix} \in \mathbb{R}^{3 \times 2}$$

$$Y = \begin{bmatrix} 1 \\ -1 \\ 2 \end{bmatrix}$$

$$w^* = (X^T \cdot X)^{-1} X^T Y = \left(\begin{bmatrix} 1 & 1 \\ 1 & 2 \\ 1 & 3 \end{bmatrix} \begin{bmatrix} 1 & 1 & 1 \\ 1 & 2 & 3 \end{bmatrix} \right)^{-1} \begin{bmatrix} 1 & 1 & 1 \\ 1 & 2 & 3 \end{bmatrix} \begin{bmatrix} 1 \\ -1 \\ 2 \end{bmatrix} =$$

$$= \begin{bmatrix} \frac{7}{3} & -1 \\ -1 & \frac{1}{2} \end{bmatrix} \begin{bmatrix} 1 & 1 & 1 \\ 1 & 2 & 3 \end{bmatrix} \begin{bmatrix} 1 \\ -1 \\ 2 \end{bmatrix} =$$

$$= \begin{pmatrix} \frac{4}{3} & \frac{1}{3} & -\frac{2}{3} \\ -\frac{1}{2} & 0 & \frac{1}{2} \end{pmatrix} \begin{pmatrix} 1 \\ -1 \\ 2 \end{pmatrix} = \begin{pmatrix} \frac{4}{3} - \frac{1}{3} - \frac{4}{3} \\ -\frac{1}{2} + 1 \end{pmatrix} = \begin{pmatrix} -\frac{1}{3} \\ \frac{1}{2} \end{pmatrix} \begin{matrix} w_0 \\ w_1 \end{matrix}$$

$$\hat{y} = w_1 x + w_0 = -\frac{1}{3} + \frac{1}{2} x$$

$$\hat{Y} = \underset{X}{\begin{pmatrix} 1 & 1 \\ 1 & 2 \\ 1 & 3 \end{pmatrix}} \underset{w}{\begin{pmatrix} -\frac{1}{3} \\ \frac{1}{2} \end{pmatrix}} = \begin{pmatrix} \frac{1}{6} \\ \frac{2}{3} \\ \frac{7}{6} \end{pmatrix}$$

$$\mathcal{L}(w) = \frac{1}{2m} (Xw - y)^T (Xw - y) = \frac{1}{6} \begin{bmatrix} \frac{1}{6} - 1 \\ \frac{2}{3} + 1 \\ \frac{7}{6} - 2 \end{bmatrix}^T \begin{bmatrix} \frac{1}{6} - 1 \\ \frac{2}{3} + 1 \\ \frac{7}{6} - 2 \end{bmatrix}$$

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③ $x^{(1)} = \begin{pmatrix} 1 \\ 0 \end{pmatrix}$ $x^{(2)} = \begin{pmatrix} -1 \\ 1 \end{pmatrix}$ $x^{(3)} = \begin{pmatrix} 0 \\ 1 \end{pmatrix}$

$y^{(1)} = 1$ $y^{(2)} = 2$ $y^{(3)} = 3$

Cerință: $w_0, w_1, w_2, \hat{y}, \mathcal{L}(w)$

$n = 3$, $d = 2$

$$X = \begin{pmatrix} 1 & 1 & 0 \\ 1 & -1 & 1 \\ 1 & 0 & 1 \end{pmatrix} \qquad y = \begin{pmatrix} 1 \\ 2 \\ 3 \end{pmatrix}$$

$$X^T \cdot X = \begin{pmatrix} 1 & 1 & 0 \\ 1 & -1 & 1 \\ 1 & 0 & 1 \end{pmatrix} \begin{pmatrix} 1 & 1 & 1 \\ 1 & -1 & 0 \\ 0 & 1 & 1 \end{pmatrix} = \begin{pmatrix} 3 & 0 & 2 \\ 0 & 2 & -1 \\ 2 & -1 & 2 \end{pmatrix}$$

$\det A = 12 - 8 - 3 = 1$

$$A^T = \begin{pmatrix} 3 & 0 & 2 \\ 0 & 2 & -1 \\ 2 & -1 & 2 \end{pmatrix} \qquad A^* = \begin{pmatrix} 3 & -2 & -4 \\ -2 & 2 & 3 \\ -4 & 3 & 6 \end{pmatrix}$$

$a_{11} = (-1)^2 \begin{vmatrix} 2 & -1 \\ -1 & 2 \end{vmatrix} = 4 - 1 = 3$ etc

$A^{-1} = A^*$

$$w^* = A^{-1} \underbrace{\begin{pmatrix} 1 & 1 & 1 \\ 1 & -1 & 0 \\ 0 & 1 & 1 \end{pmatrix}}_{X^T} \underbrace{\begin{pmatrix} 1 \\ 2 \\ 3 \end{pmatrix}}_{y} = \begin{pmatrix} 0 \\ 1 \\ 3 \end{pmatrix} \begin{matrix} w_0 \\ w_1 \\ w_2 \end{matrix}$$

$\hat{y} = 0 + 1 \cdot x_1 + 3 x_2$

$$\hat{y} = X w^* = \begin{pmatrix} 1 & 1 & 0 \\ 1 & -1 & 1 \\ 1 & 0 & 1 \end{pmatrix} \begin{pmatrix} 0 \\ 1 \\ 3 \end{pmatrix} = \begin{pmatrix} 1 \\ 2 \\ 3 \end{pmatrix} \quad \hat{y} = y$$

$\Rightarrow$ nu ar trb să existe loss

$$\mathcal{L}(w) = \frac{1}{2m}\underbrace{(Xw-y)}^{T}(Xw-y) = \frac{1}{2m}(\hat{y}-y)^T(\hat{y}-y) =$$

$$= \frac{1}{2m}\cdot(0\ \ 0\ 0)\begin{pmatrix}0\\0\\0\end{pmatrix} = \frac{1}{6}\cdot 0 = 0 \quad \checkmark$$

4.

Question 1
Not yet answered
Marked out of 1.00
Flag question

Assume we have a dataset $\{(x^{(1)}, y^{(1)}), (x^{(2)}, y^{(2)}), \ldots, (x^{(n)}, y^{(n)})\}$ of points in the 2D plane, and we want to estimate the following linear regression model:

$\hat{y} = w_1 x + w_0$.

Write the batch gradient descent update for w_0, w_1 in terms of $x^{(1)}, y^{(1)}, x^{(2)}, y^{(2)}, \ldots, x^{(n)}, y^{(n)}$, where the learning rate is η. Numerical application: $(1,1), (2,-1), (3,2)$, $\eta = 0.1$.

Formula: $w \longleftarrow w - \eta \nabla_w \mathcal{L}(w)$

$\nabla_w \mathcal{L}(w) = \frac{1}{n} X^T (Xw - y)$

Rezolvare: $n = 3$, $d = 1$

$$X = \begin{pmatrix} 1 & 1 \\ 1 & 2 \\ 1 & 3 \end{pmatrix} \quad y = \begin{pmatrix} 1 \\ -1 \\ 2 \end{pmatrix} \quad X^T = \begin{pmatrix} 1 & 1 & 1 \\ 1 & 2 & 3 \end{pmatrix} \quad w = \begin{pmatrix} w_0 \\ w_1 \end{pmatrix} \in \mathbb{R}^{(d+1)\times 1} = \mathbb{R}^{2\times 1}$$

$$Xw - y = \begin{pmatrix} 1 & 1 \\ 1 & 2 \\ 1 & 3 \end{pmatrix} \begin{pmatrix} w_0 \\ w_1 \end{pmatrix} - \begin{pmatrix} 1 \\ -1 \\ 2 \end{pmatrix} = \begin{pmatrix} w_0 + w_1 - 1 \\ w_0 + 2w_1 + 1 \\ w_0 + 3w_1 - 2 \end{pmatrix}$$

$$\frac{1}{n} X^T (Xw - y) = \frac{1}{3} \begin{pmatrix} 1 & 1 & 1 \\ 1 & 2 & 3 \end{pmatrix} \begin{pmatrix} w_0 + w_1 - 1 \\ w_0 + 2w_1 + 1 \\ w_0 + 3w_1 - 2 \end{pmatrix} = \frac{1}{3} \begin{pmatrix} 3w_0 + 6w_1 - 2 \\ 6w_0 + 14w_1 - 5 \end{pmatrix}$$

$$w \longleftarrow \begin{pmatrix} w_0 \\ w_1 \end{pmatrix} - \frac{0,1}{3} \begin{pmatrix} 3w_0 + 6w_1 - 2 \\ 6w_0 + 14w_1 - 5 \end{pmatrix}$$

Despărțim pe fiecare termen

$$w_0 \longleftarrow 0,9\, w_0 - 0,2\, w_1 + \frac{0,2}{3}$$

$$w_1 \longleftarrow -2\, w_0 + \frac{1,6}{3} w_1 + \frac{0,5}{3}$$

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5. $x^{(1)} = \begin{pmatrix} 1 \\ 0 \end{pmatrix}$ $x^{(2)} = \begin{pmatrix} -1 \\ 1 \end{pmatrix}$ $x^{(3)} = \begin{pmatrix} 0 \\ 1 \end{pmatrix}$ $n = 3$, $d = 2$

$y^{(1)} = 1$ $y^{(2)} = 2$ $y^{(3)} = 3$

$w^0 = \begin{pmatrix} 1 \\ -1 \\ 3 \end{pmatrix}$, $\eta = 0,1$ (valoare aleatorie)

$$X = \begin{pmatrix} 1 & 1 & 0 \\ 1 & -1 & 1 \\ 1 & 0 & 1 \end{pmatrix} \qquad Y = \begin{pmatrix} 1 \\ 2 \\ 3 \end{pmatrix}$$

$$w^1 \leftarrow w^0 - \eta \nabla_w \mathcal{L}(w^0)$$

$$\nabla_w \mathcal{L}(w^0) = \frac{1}{n} X^T (Xw - Y) =$$

$$= \frac{1}{3} \begin{pmatrix} 1 & 1 & 1 \\ 1 & -1 & 0 \\ 0 & 1 & 1 \end{pmatrix} \left[\begin{pmatrix} 1 & 1 & 0 \\ 1 & -1 & 1 \\ 1 & 0 & 1 \end{pmatrix} \begin{pmatrix} 1 \\ -1 \\ 3 \end{pmatrix} - \begin{pmatrix} 1 \\ 2 \\ 3 \end{pmatrix} \right] =$$

$$= \frac{1}{3} \begin{pmatrix} 1 & 1 & 1 \\ 1 & -1 & 0 \\ 0 & 1 & 1 \end{pmatrix} \left[\begin{pmatrix} 0 \\ 4 \\ 3 \end{pmatrix} - \begin{pmatrix} 1 \\ 2 \\ 3 \end{pmatrix} \right] = \frac{1}{3} \begin{pmatrix} 1 & 1 & 1 \\ 1 & -1 & 0 \\ 0 & 1 & 1 \end{pmatrix} \begin{pmatrix} -1 \\ 2 \\ 0 \end{pmatrix} =$$

$$= \frac{1}{3} \begin{pmatrix} 1 \\ -3 \\ 2 \end{pmatrix}$$

$$w_1 \leftarrow \begin{pmatrix} 1 \\ -1 \\ 3 \end{pmatrix} - \frac{0,1}{3} \begin{pmatrix} 1 \\ -3 \\ 2 \end{pmatrix} = \begin{pmatrix} 29/30 \\ -9/30 \\ 58/30 \end{pmatrix}$$

6.

Rândul 5

1. Assume we have a dataset $\{(x^{(1)}, y^{(1)}), (x^{(2)}, y^{(2)}), (x^{(3)}, y^{(3)})\} = \{(1,1), (2,-1), (3,2)\}$ of points in the 2D plane, and we want to estimate the following ridge regression model with hyperparameter $\lambda = 2$:

$$\hat{y} = w_1 x + w_0.$$

Compute w_0, w_1 and $\hat{y}^{(1)}, \hat{y}^{(2)}, \hat{y}^{(3)}$.

Formule: $\mathcal{L}(w) = \frac{1}{2n}(Xw - y)^T(Xw - y) + \frac{1}{2}\lambda \|w\|_2^2$

$$\|w\|_2 = \sqrt{\sum_{j=0}^{d} |w_j|^2}$$

$$w^* = (X^T X + \lambda I)^{-1} X^T y$$

Rezolvare: $n = 3$ $d = 1$ $\lambda = 2$

$$X = \begin{pmatrix} 1 & 1 \\ 1 & 2 \\ 1 & 3 \end{pmatrix} \quad y = \begin{pmatrix} 1 \\ -1 \\ 2 \end{pmatrix} \quad X^T = \begin{pmatrix} 1 & 1 & 1 \\ 1 & 2 & 3 \end{pmatrix}$$

$$X^T X + \lambda I = \begin{pmatrix} 1 & 1 \\ 1 & 2 \\ 1 & 3 \end{pmatrix} \begin{pmatrix} 1 & 1 & 1 \\ 1 & 2 & 3 \end{pmatrix} + 2\begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} = \begin{pmatrix} 3 & 6 \\ 6 & 14 \end{pmatrix} + \begin{pmatrix} 2 & 0 \\ 0 & 2 \end{pmatrix} =$$

$$= \begin{pmatrix} 5 & 6 \\ 6 & 16 \end{pmatrix} \qquad \det(X^T X + \lambda I) = 80 - 36 = 44$$

$$(X^T X + \lambda I)^{-1} = \frac{1}{44} \cdot \begin{pmatrix} 16 & -6 \\ -6 & 5 \end{pmatrix}$$

$$w^* = \frac{1}{44} \cdot \begin{pmatrix} 16 & -6 \\ -6 & 5 \end{pmatrix} \begin{pmatrix} 1 & 1 & 1 \\ 1 & 2 & 3 \end{pmatrix} \begin{pmatrix} 1 \\ -1 \\ 2 \end{pmatrix} =$$

$$= \frac{1}{44} \begin{pmatrix} 10 & 4 & -2 \\ -1 & 4 & 9 \end{pmatrix} \begin{pmatrix} 1 \\ -1 \\ 2 \end{pmatrix} = \frac{1}{44} \begin{pmatrix} 2 \\ 13 \end{pmatrix}$$

$$\Rightarrow w_0 = \frac{1}{22} \text{ și } w_1 = \frac{13}{44}$$

$$\hat{y} = \frac{1}{21} + \frac{13}{44} x$$

$$\hat{y} = X w = \begin{pmatrix} 1 & 1 \\ 1 & 2 \\ 1 & 3 \end{pmatrix} \begin{pmatrix} 0,04 \\ 0,3 \end{pmatrix} = \begin{pmatrix} 0,34 \\ 0,64 \\ 0,94 \end{pmatrix} \quad \begin{matrix} \hat{y}^{(1)} \\ \hat{y}^{(2)} \\ \hat{y}^{(3)} \end{matrix}$$

Dacă ar cere și loss

$$\mathcal{L}(w) = \frac{1}{2 \cdot 3} \begin{pmatrix} 0,34 & -1 \\ 0,64 & +1 \\ 0,94 & -2 \end{pmatrix}^T \cdot \begin{pmatrix} 0,34-1 \\ 0,64+1 \\ 0,94-2 \end{pmatrix} + \frac{1}{2} \cdot 2 \, \| w \|_2^2$$

$$\| w \|_2 = \sqrt{w_0^2 + w_1^2}$$

7.

Rândul 4

1. Assume we have a dataset $\{(x^{(1)}, y^{(1)}), (x^{(2)}, y^{(2)}), \ldots, (x^{(n)}, y^{(n)})\}$ of points in the 2D plane, and we want to estimate the following elastic net regression model with hyperparameters λ and r:

$$\hat{y} = w_1 x + w_0,$$

Write the batch gradient descent update for w_0, w_1 in terms of $x^{(1)}, y^{(1)}, x^{(2)}, y^{(2)}, \ldots, x^{(n)}, y^{(n)}, \lambda, r$, where the learning rate is η. Numerical application: $\{(x^{(1)}, y^{(1)}), (x^{(2)}, y^{(2)}), (x^{(3)}, y^{(3)})\} = \{(1,1), (2,-1), (3,2)\}$, $\eta = 0.1$, $\lambda = 1$, $r = 0.5$.

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Formule: $\mathcal{L}(w) = \frac{1}{2n} (Xw - y)^T (Xw - y) + r\lambda \|w\|_1 + \frac{1-r}{2} \lambda \|w\|_2^2$

$\|w\|_1 = \sum_{j=0}^{d} |w_j| \longrightarrow \text{sign}(w_j)$

$\|w\|_2 = \sqrt{\sum_{j=1}^{d} |w_j|^2}$ $\qquad \frac{\partial \mathcal{L}}{\partial w}$

$w \longleftarrow w - \eta \nabla_w \mathcal{L}(w)$

Rezolvare:

$\nabla_w \mathcal{L}(w) = \frac{1}{n} X^T (Xw - y) + r\lambda \, \text{sign}(w) + (1-r)\lambda w$

$\frac{\partial \|w\|_1}{\partial w} = \text{sign}(w)$ $\qquad \frac{\partial \|w\|_2^2}{\partial w} = 2w$

$n = 3, \; d = 1$

$$X = \begin{pmatrix} 1 & 1 \\ 1 & 2 \\ 1 & 3 \end{pmatrix} \qquad y = \begin{pmatrix} 1 \\ -1 \\ 2 \end{pmatrix}$$

$$Xw - y = \begin{pmatrix} 1 & 1 \\ 1 & 2 \\ 1 & 3 \end{pmatrix} \begin{pmatrix} w_0 \\ w_1 \end{pmatrix} - \begin{pmatrix} 1 \\ -1 \\ 2 \end{pmatrix} = \begin{pmatrix} w_0 + w_1 - 1 \\ w_0 + 2w_1 + 1 \\ w_0 + 3w_1 - 2 \end{pmatrix}$$

$$X^T (Xw - y) = \begin{pmatrix} 1 & 1 & 1 \\ 1 & 2 & 3 \end{pmatrix} \begin{pmatrix} w_0 + w_1 - 1 \\ w_0 + 2w_1 + 1 \\ w_0 + 3w_1 - 2 \end{pmatrix} = \begin{pmatrix} 3w_0 + 6w_1 - 2 \\ 6w_0 + 14w_1 - 5 \end{pmatrix}$$

$$\nabla_w \mathcal{L}(w) = \frac{1}{3} \begin{pmatrix} 3w_0 + 6w_1 - 2 \\ 6w_0 + 14w_1 - 5 \end{pmatrix} + 0{,}5 \cdot 1 \begin{pmatrix} \text{sign}(w_0) \\ \text{sign}(w_1) \end{pmatrix} + (1 - 0{,}5) \cdot 1 \begin{pmatrix} w_0 \\ w_1 \end{pmatrix} =$$

$$= \begin{pmatrix} w_0 + 2w_1 - \frac{2}{3} + 0{,}5\,\mathrm{sign}(w_0) + 0{,}5\,w_0 \\ 2w_0 + \frac{14}{3}w_1 - \frac{5}{3} + 0{,}5\,\mathrm{sign}(w_1) + 0{,}5 \cdot w_1 \end{pmatrix}.$$

$$w^{(1)} \longleftarrow \begin{pmatrix} w_0 \\ w_1 \end{pmatrix} - 0{,}1 \cdot \begin{pmatrix} w_0 + 2w_1 - \frac{2}{3} + 0{,}5\,\mathrm{sign}(w_0) + 0{,}5\,w_0 \\ 2w_0 + \frac{14}{3}w_1 - \frac{5}{3} + 0{,}5\,\mathrm{sign}(w_1) + 0{,}5 \cdot w_1 \end{pmatrix}$$

8.

1. Assume we have a dataset $\{(x_1^{(1)}, x_2^{(1)}, y^{(1)}), (x_1^{(2)}, x_2^{(2)}, y^{(2)}), (x_1^{(3)}, x_2^{(3)}, y^{(3)}), (x_1^{(4)}, x_2^{(4)}, y^{(4)})\} = \{(1,2,1), (2,3,-1), (3,4,2), (4,5,3)\}$ of points in 3D, and we want to estimate the following elastic net regression model with hyperparameters $\lambda = 4$ and $r = 0.75$:

$$\hat{y} = w_2 x_2 + w_1 x_1 + w_0,$$

Write the batch gradient descent update for w_0, w_1, w_2, where the learning rate is $\eta = 0.4$.

Formule: $\mathcal{L}(w) = \frac{1}{2n} (Xw - y)^T (Xw - y) + r\lambda \|w\|_1 + \frac{1-r}{2} \lambda \|w\|_2^2$

$\|w\|_1 = \sum_{j=0}^{d} |w_j| \longrightarrow \text{sign}(w_j)$

$\|w\|_2 = \sqrt{\sum_{j=1}^{d} |w_j|^2}$ $\qquad \frac{\partial \mathcal{L}}{\partial w}$

$w \longleftarrow w - \eta \nabla_w \mathcal{L}(w)$

Rezolvare: $n = 4$, $d = 2$

$$X = \begin{pmatrix} 1 & 1 & 2 \\ 1 & 2 & 3 \\ 1 & 3 & 4 \\ 1 & 4 & 5 \end{pmatrix} \qquad y = \begin{pmatrix} 1 \\ -1 \\ 2 \\ 3 \end{pmatrix}$$

$$\nabla_w \mathcal{L}(w) = \frac{1}{n} X^T (Xw - y) + r\lambda \, \text{sign}(w) + (1-r)\lambda w$$

$$\frac{\partial \|w\|_1}{\partial w} = \text{sign}(w) \qquad \frac{\partial \|w\|_2^2}{\partial w} = 2w$$

$$Xw - y = \begin{pmatrix} 1 & 1 & 2 \\ 1 & 2 & 3 \\ 1 & 3 & 4 \\ 1 & 4 & 5 \end{pmatrix} \begin{pmatrix} w_0 \\ w_1 \\ w_2 \end{pmatrix} - \begin{pmatrix} 1 \\ -1 \\ 2 \\ 3 \end{pmatrix} = \begin{pmatrix} w_0 + w_1 + 2w_2 - 1 \\ w_0 + 2w_1 + 3w_2 + 1 \\ w_0 + 3w_1 + 4w_2 - 2 \\ w_0 + 4w_1 + 5w_2 - 3 \end{pmatrix}$$

$$X^T (Xw - y) = \begin{pmatrix} 1 & 1 & 1 & 1 \\ 1 & 2 & 3 & 4 \\ 2 & 3 & 4 & 5 \end{pmatrix} \begin{pmatrix} w_0 + w_1 + 2w_2 - 1 \\ w_0 + 2w_1 + 3w_2 + 1 \\ w_0 + 3w_1 + 4w_2 - 2 \\ w_0 + 4w_1 + 5w_2 - 3 \end{pmatrix} =$$

$$= \begin{pmatrix} 4w_0 + 10w_1 + 14w_2 - 5 \\ 10w_0 + 30w_1 + 40w_2 - 17 \\ 15w_0 + 40w_1 + 54w_2 - 22 \end{pmatrix}$$

$$\nabla_w \mathcal{L}(w) = \frac{1}{4}\begin{pmatrix} 4w_0 + 10w_1 + 14w_2 - 5 \\ 10w_0 + 30w_1 + 40w_2 - 17 \\ 15w_0 + 40w_1 + 54w_2 - 22 \end{pmatrix} + 0{,}75 \cdot 4 \begin{pmatrix} \operatorname{sign}(w_0) \\ \operatorname{sign}(w_1) \\ \operatorname{sign}(w_2) \end{pmatrix} + (1-0{,}75)\begin{pmatrix} w_0 \\ w_1 \\ w_2 \end{pmatrix} =$$

$$= \begin{pmatrix} w_0 + \frac{5}{2}w_1 + \frac{7}{2}w_2 - \frac{5}{4} + 3\cdot\operatorname{sign}(w_0) + 0{,}25\, w_0 \\ \frac{5}{2}w_0 + \frac{15}{2}w_1 + 10w_2 - \frac{17}{4} + 3\operatorname{sign}(w_1) + 0{,}25\, w_1 \\ \frac{15}{4}w_0 + 10w_1 + \frac{27}{2}w_2 - \frac{11}{2} + 3\operatorname{sign}(w_2) + 0{,}25\, w_2 \end{pmatrix}$$

$$w \longleftarrow \begin{pmatrix} w_0 \\ w_1 \\ w_2 \end{pmatrix} - 0{,}4 \begin{pmatrix} w_0 + \frac{5}{2}w_1 + \frac{7}{2}w_2 - \frac{5}{4} + 3\cdot\operatorname{sign}(w_0) + 0{,}25\, w_0 \\ \frac{5}{2}w_0 + \frac{15}{2}w_1 + 10w_2 - \frac{17}{4} + 3\operatorname{sign}(w_1) + 0{,}25\, w_1 \\ \frac{15}{4}w_0 + 10w_1 + \frac{27}{2}w_2 - \frac{11}{2} + 3\operatorname{sign}(w_2) + 0{,}25\, w_2 \end{pmatrix}$$